THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The unit of electric power is: 1

(a) joule **(b)** ohm **(c)** watt **(d)** volt

2. A student states: “Nichrome has higher resistivity than copper.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) They are equal **(d)** Cannot be determined

3. Resistance of a 2 m wire is 4 Ω. A 6 m wire of the same material and thickness has resistance: 1

(a) 24 Ω **(b)** 12 Ω **(c)** 8 Ω **(d)** 4/3 Ω

4. The charge on one electron is: 1

(a) 1.6×10^{19} C **(b)** 9.1×10^{-31} C **(c)** 1.6×10^{-19} C **(d)** 1 C

5. For two 4 Ω resistors in series, Student A says 8 Ω; Student B says 2 Ω. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. Which instrument detects a magnetic field? 1

(a) compass **(b)** rheostat **(c)** ammeter **(d)** voltmeter

7. A student states: “The force on a conductor becomes zero if it is parallel to the magnetic field.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Only for AC **(d)** Cannot be determined

8. In a domestic circuit, appliances are connected: 1

(a) randomly **(b)** in series **(c)** in both **(d)** in parallel

9. The rating of a fuse used for a 2 kW geyser at 220 V is about: 1

(a) 50 A **(b)** 10 A **(c)** 5 A **(d)** 1 A

10. For the field at the centre of a circular coil, Student A says it depends on the radius; Student B says it does not. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. How much current flows through a 2 kW heater at 220 V? 2

12. What is the function of a fuse in a circuit? 2

13. What is the function of an MCB? 2

14. What does the crowding of field lines indicate? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A nichrome wire has resistance 20 Ω. Find the resistance of another nichrome wire of the same length but double the area. 3

16. What are the colour codes of live, neutral and earth wires? Why is the switch connected in the live wire? 3

17. Compare the power used in a 2 Ω resistor in each case: 3
(i) 6 V battery in series with 1 Ω and 2 Ω
(ii) 4 V battery in parallel with 12 Ω and 2 Ω.

OR

What are the applications of the heating effect of electric current? (three) 3

18. What happens to the force on a conductor when 3
(i) current is increased
(ii) field is reversed
(iii) the conductor is made parallel to field?

OR

Distinguish between short-circuiting and overloading. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A student measured the resistance of wires A and B of the same material. A is 1 m long, 1 mm² area; B is 2 m long, 0.5 mm² area. 5
(i) Which has more resistance?
(ii) Find the ratio R_B / R_A.
(iii) What is resistivity?
(iv) Why is resistivity the same for both?
(v) Name a material with very low resistivity.

OR

(i) How many resistors of 10 Ω each are needed to carry 2.5 A on a 5 V line when connected in parallel? 5
(ii) Show the combination of four 4 Ω resistors giving 4 Ω.

20. **Case study:** In an experiment, an aluminium rod suspended between the poles of a magnet moves when current is passed. 5
(i) Name the phenomenon.
(ii) If the field is vertically upward and the current flows from west to east, find the force direction.
(iii) What happens if both current and field are reversed?
(iv) Name the rule used.
(v) Why is aluminium preferred?

OR

(i) What is the right-hand thumb rule? 5
(ii) Apply it to find the field direction inside a circular loop carrying anticlockwise current (as seen from the front).
(iii) How do field lines at the centre look?

END OF QUESTION PAPER — Total: 40 Marks

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